

CareFlow: Smart Appointment & Lead Management System

A major capstone project built during a Web Development Internship with LaunchED, CareFlow is a full-stack healthcare management platform designed to streamline appointments, lead tracking, follow-ups, and patient engagement for small clinics and coaching centres.

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The Problem: Manual Chaos in Healthcare

Small clinics rely on WhatsApp, spreadsheets, and manual registers. This leads to missed appointments, weak follow-ups, poor lead tracking, manual errors, and revenue leakage.



The Solution: CareFlow Platform

Lead Management

**Patient
Management**

**Appointment
Scheduling**

**Treatment
Tracking**

**Payment
Management**

**Treatment
Progress Logs**

**Reports &
Analytics
Dashboard**

**JWT
Authentication &
Secure Access**



Security & Reliability First

Secure Passwords

Passwords are securely hashed using bcrypt.

Protected Access

JWT-based authentication and protected routes keep access controlled.

Private Data

Sensitive user information is hidden from API responses.

Admin Safeguard

Prevents deletion of the last admin account.

- ❑ Primary audience: clinic owners and coaching centre managers evaluating web solutions; useful for technical stakeholders and small-business investors.

Tech Stack

Frontend

React.js (Vite), Tailwind CSS, JavaScript, Axios, Recharts.

Backend

Node.js, Express.js.

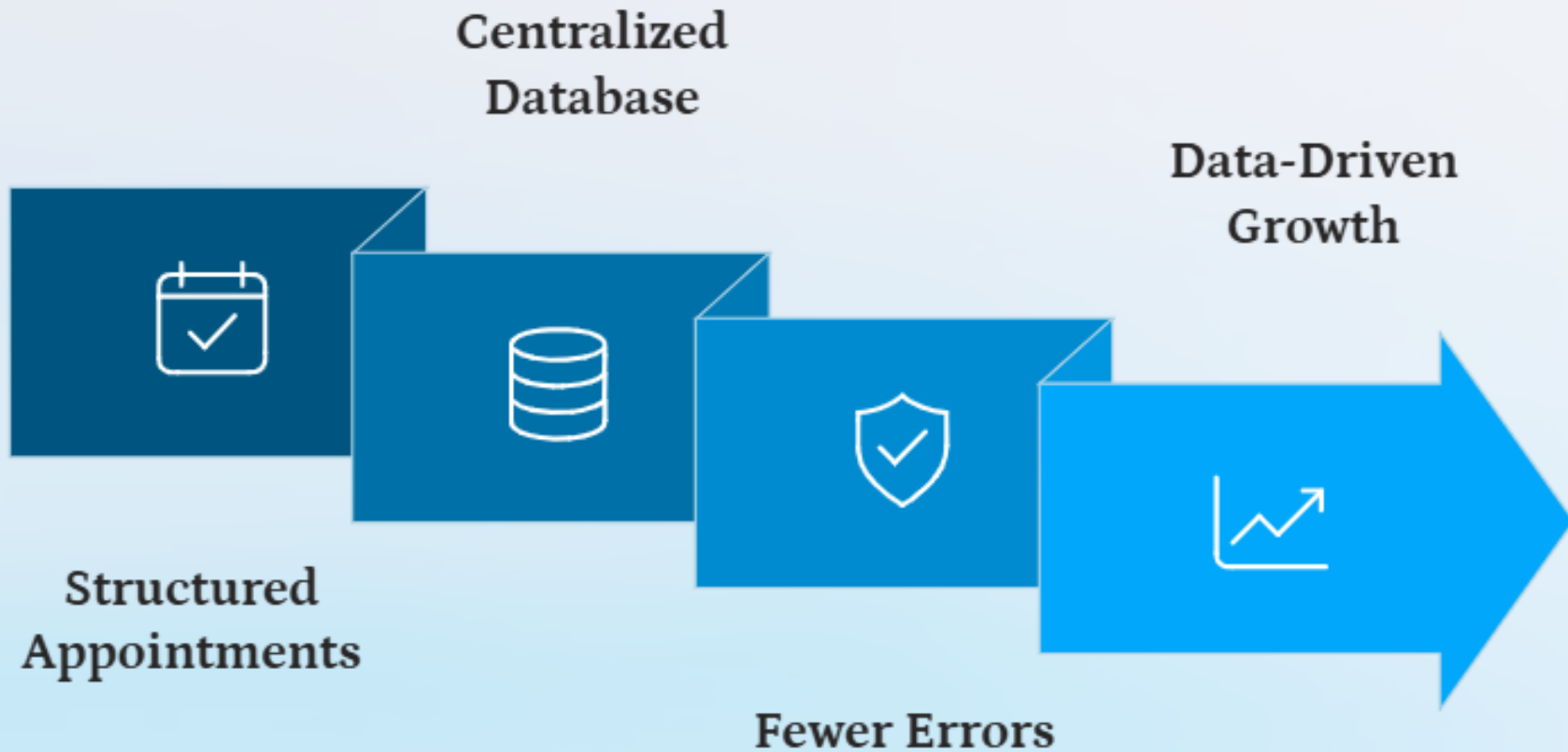
Database

Supabase PostgreSQL.

Deployment

Vercel, Render.

Outcomes & Impact



Skills Gained & Applied



Authentication & Authorization

Protect access with secure sign-in, role-based permissions, and controlled data visibility.



Full Stack Development

Build end-to-end features across the UI, APIs, and database layers for a seamless experience.



Database Design & Management

Structure relational data for reliable storage, clean querying, and scalable record management.



Debugging & Problem Solving

Diagnose issues quickly, trace root causes, and implement reliable fixes across the stack.

Skills Gained & Applied



API Integration

Connect SMS/email providers and analytics endpoints to automate communications and track conversions.



CRUD Operations

Lead lifecycle management: create, read, update, delete records securely with role-based access.



Conversion Optimization

Improve user flows and calls to action to increase form completion and appointment bookings.



Form Handling

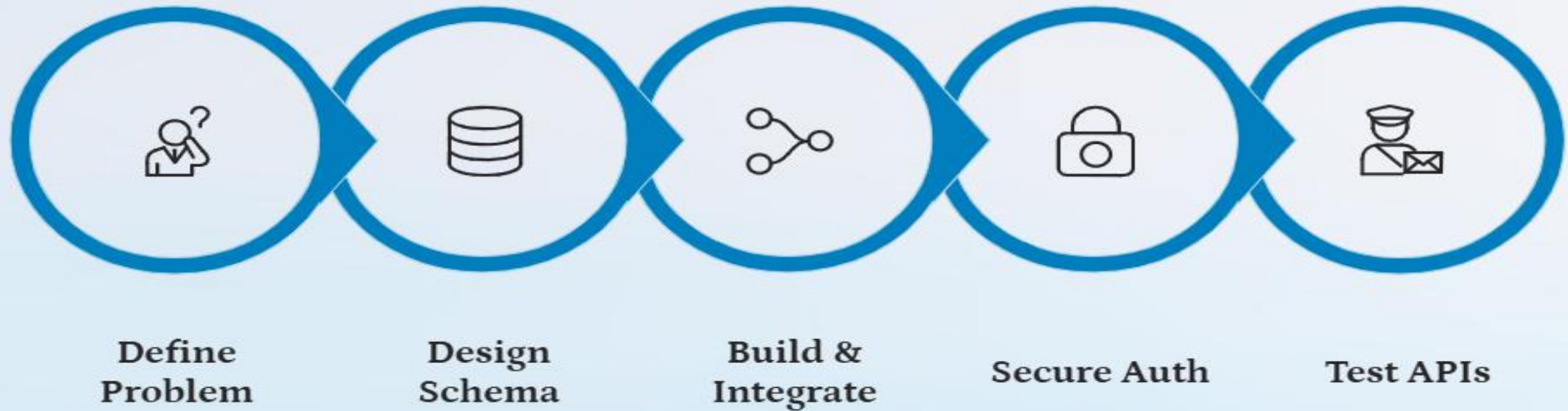
Robust client-side validation with server-side checks to ensure accurate lead capture and reduce spam.



Deployment & Cloud Hosting

CI/CD using GitHub + Vercel for rapid releases and zero-downtime updates.

The Learning Journey



This project allowed me to experience the complete software development lifecycle—from understanding the problem statement, designing the database schema and system architecture, building and integrating frontend and backend modules, implementing secure authentication, testing every API endpoint with Postman, resolving database constraints, debugging routing and deployment issues, and finally deploying, documenting, and maintaining a production-ready application.

I also used AI as a learning partner throughout the process, but critical thinking and hands-on problem-solving were essential to making progress and ensuring the final solution was reliable.

Live Demo & Code

Live Demo

<https://careflow-green.vercel.app>

GitHub Repository

<https://github.com/santhipriyan2007/careflow>